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ITAI 2372

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An Analysis of AI Use Cases in Healthcare

❖ Industry overview

- The healthcare industry focuses mostly on preventing, diagnosing, and treating a variety of illnesses, while also helping people maintain and improve their overall health. It aids and assists patients, doctors, nurses, hospitals, clinics, researchers, and many other healthcare professionals. Healthcare is one of the most important industries in the world because everyone depends on it at some point in their lives.
- Today, healthcare faces several challenges, including rising costs, increasing patient populations, and the need for faster and more accurate diagnoses. This is where artificial intelligence (AI) is making an impactful difference. AI helps healthcare organizations improve patient care, streamline operations, and support medical research. As technology continues to both improve and advance, AI is becoming an increasingly important tool in modern healthcare.
- As healthcare continues to change, technology is becoming more important than ever. AI has the potential to help healthcare providers improve patient care while also making many processes faster and more efficient.

❖ Current AI trends

- One of the most common and prevalent uses of AI in healthcare today is medical diagnosis. AI systems can analyze medical images such as X-rays, MRIs, and CT scans to help doctors identify diseases and abnormalities. These tools can process large amounts of information quickly and help healthcare professionals detect conditions much earlier. This can result in faster treatment and better patient outcomes.
- Another growing AI trend is personalized medicine. AI can analyze a patient's medical history, lifestyle, and genetic information to help doctors create treatment plans that are tailored to the individual. Instead of using the same treatment plan

for every patient, healthcare providers can use AI to create more personalized treatment plans that may improve effectiveness and reduce side effects.

- AI is also being used in drug discovery and development. Developing new medications can take years and cost billions of dollars. AI helps researchers analyze data more efficiently and identify promising drug options faster than traditional methods. A well-known example of this is Moderna, which used AI to help accelerate the development and rollout of its COVID-19 vaccine. By speeding up multiple parts of the research and development process, AI helped support one of the most important medical breakthroughs of recent years. These examples show that AI is already making a significant impact across multiple areas of healthcare. As technology continues to improve, healthcare organizations will likely find even more ways to use AI to support patients and medical professionals.
- AI is also helping hospitals improve their day-to-day operations. Many healthcare organizations use AI to help manage schedules, allocate resources, and reduce administrative work. This allows doctors and healthcare staff to spend more time focusing on patient care instead of paperwork and routine tasks.

❖ Ethical implications

- Although AI offers many benefits, it also raises important ethical concerns. One major concern is privacy and data security. Healthcare AI systems rely on large amounts of patient information, including medical records and personal health data. If this information is not properly protected, patients could face privacy violations or data breaches. Healthcare organizations must ensure that sensitive information is stored and used responsibly.
- Another concern is bias and fairness. AI systems learn from data, and if the data used to train them is incomplete or biased, the results may not be equally accurate for all patients. This could lead to unfair treatment or misdiagnosis for certain groups of people. Developers and healthcare providers must work to make sure AI systems are trained using diverse and representative data.
- Transparency is another important issue. Doctors and patients need to understand how AI systems make recommendations and decisions. If an AI system provides a diagnosis without explaining how it reached that conclusion, it may be difficult for healthcare professionals to trust its results. AI should support medical professionals rather than replace human judgment entirely. Addressing these concerns will be critical as AI becomes more common in healthcare settings. Building trust between patients, healthcare providers, and AI systems will be essential for successful adoption in the future.

- Another concern is that people may become too dependent on AI systems. While AI can provide helpful recommendations and insights, it should not replace the experience and judgment of healthcare professionals. Doctors still play an important role in reviewing information, making final decisions, and providing care that takes a patient's unique situation into account.

❖ Future trends

- Over the next five to ten years, AI is expected to become even more integrated into healthcare. One future trend is the growth of AI-powered virtual health assistants. These systems may be able to answer patient questions, monitor health conditions, and provide personalized health recommendations. Combined with telemedicine, AI could make healthcare more accessible, especially for people who live far from hospitals or healthcare providers.
- Another exciting development is predictive healthcare. AI may be able to identify diseases before symptoms become noticeable by analyzing health records, wearable device data, and genetic information. Earlier detection could allow doctors to begin treatment sooner and improve patient outcomes.
- AI is also expected to play a larger role in personalized medicine and genomics. Researchers are using AI to better understand how genetics affect health and disease. In the future, treatments may become even more customized based on an individual's unique genetic profile.
- Another area that is likely to grow is the use of AI in robotic surgery. As technology continues to improve, AI-assisted surgical systems may help doctors perform procedures with even greater precision. This could lead to fewer complications, shorter recovery times, and better outcomes for patients.
- While these advancements offer many opportunities, there are also risks to consider. Healthcare organizations will need to address issues such as cybersecurity, patient privacy, and the possibility of becoming too dependent on AI systems. As AI continues to grow, strong regulations and ethical guidelines will be important to ensure that these technologies are used safely and responsibly.

❖ Personal reflection

- What surprised me most about this topic was how much AI is already being used in healthcare today. Before learning about this module, I mostly thought of AI as

chatbots, virtual assistants, or tools used for writing and searching for information. I was surprised to learn that AI is already helping doctors analyze medical images, support surgeries, discover new drugs, and create personalized treatment plans for patients.

- Learning about AI in healthcare also changed my perspective on how technology can be used to solve real-world problems. It showed me that AI is not just about making tasks easier; it can also help improve patient care and potentially save lives.
- I would consider working in the healthcare AI industry because it combines technology with the opportunity to make a meaningful impact on people's lives. The idea of using AI to help doctors make better decisions and improve patient outcomes is exciting to me. At the same time, I believe it is important to address concerns such as privacy, bias, and transparency to ensure AI is used fairly and responsibly. Overall, I think AI has the potential to make healthcare more effective, accessible, and efficient in the future.

❖ References

- ITAI 2372 Module 03 Lecture Slides: AI in Healthcare and Agriculture.
- Moderna. (2024). AI and vaccine development.
- World Health Organization. (2024). Ethics and governance of artificial intelligence for health.